Racing Genesis: Production Planning & MVP 2.0 Alpha Release Report

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Project Links & Verification Access

Live Hosted URL: Racing Genesis - F1 Pro Simulator

Local Offline Execution: cd Assignment-4/prototype && python serve.py 8081 → http://localhost:8081

Production Schedule & Weekly Development Plan (Weeks 9–14)

The remaining semester development roadmap establishes engineering, asset creation, and system integration targets leading to an uncompromised final buffer week:

Week	Milestone & Focus	Engineering Deliverables	Asset & Environment Deliverables	Integration Target
Week 9 (Current)	MVP 2.0 (Alpha Build) Interactive Game Loop	Complete 3-state loop (Start → Active → End); in-race Pause modal (Esc); Web Audio engine; 90 FPS locked delta cap; manual WASD driving with reverse.	Open-cockpit 3D steering wheel dashboard with active shift lights; 5 Grand Prix presets (Monza, Silverstone, Spa, Monaco, Suzuka); dynamic mountains and forest rings.	Playable Alpha Build published on GitHub Pages and verified at 90 FPS.
Week 10	Visual Shaders & Particles	GPU particle instancing for dynamic tire smoke under threshold braking and high slip angles; asphalt skidmark decal projection; brake rotor incandescence shader.	Procedural skybox cubemap atmosphere; dynamic asphalt normal mapping and rubber accumulation racing line.	Visual FX subsystem merged into primary WebGL rendering pipeline.
Week 11	Rigid Collisions & Limits	Curvilinear road boundary collision response with elastic impulse deflection; curb rumble strip haptic/vibration feedback; off-track grass/gravel deceleration penalties.	3D Tecpro barrier impact geometry, gravel particle displacement, and track distance marker boards.	Boundary collision subsystem integrated with chassis yaw inertia.
Week 12	Multi-Agent AI Racing	Multi-car traffic simulation with competitive autonomous opponents; behavior-tree overtaking logic and lane defense; slipstream / DRS aerodynamic draft drag reduction.	Distinct team liveries for opponent vehicles; real-time race position tracker and mini-map visualizer.	Autonomous opponent racing grid functioning with zero collision deadlocks.
Week 13	Telemetry & Beta RC	Canvas-based live telemetry graphing (speed, lateral G, throttle, brake); CSV/JSON session log exporter; WebGL 2.0 draw call batching and memory optimization.	Final multi-channel audio balance (engine harmonics, wind noise, tire screech, transmission gear whine).	Beta Release Candidate locked and distributed for feedback.
Week 14 (BUFFER)	MANDATORY BUFFER WEEK Zero Feature Freeze	STRICT FEATURE FREEZE: Absolutely zero new gameplay mechanics planned. Dedicated exclusively to garbage collection profiling, memory leak audits, and cross-browser validation (Chrome, Edge, Firefox, Safari).	Final 4K promotional media renders, presentation deck preparation, and comprehensive final project report.	Final Golden Master Build & Live Project Presentation.

MVP 2.0 Alpha Release Notes & Prototype Comparison

System Evolution: Assignment 3 Midterm MVP vs. Assignment 4 Alpha

Core Dimension	Midterm Prototype (Assignment 3)	Latest Alpha Build (Assignment 4)
Architecture & Scope	Single-state technical proof-of-concept (3,089 lines); basic lap counter without menus or end states.	Complete 3-State cohesive product (4,261 lines); Start Menu circuit picker, in-race Pause modal, and Checkered Flag podium screen.
Driving Controls	Autonomous AI spline-following only; no player control over steering, throttle, or braking.	Dual-Mode system (Manual WASD + AI Autopilot); full throttle (W), threshold braking/reverse to -18 m/s (S), and responsive steering (A/D).
Track Scalability	Static single track geometry (~5,200m) with fixed environment bounds.	Dynamic scaling from 1,000m to 10,000m with live 3D anchor manipulation and 5 authentic Grand Prix presets (Monza, Spa, Monaco, etc.).
Environment & Audio	Static flat ground plane; no sound synthesis or spectator structures.	Dynamic mountain ranges, adaptive pine forest belts, clearance-checked grandstands (>=6m buffer), and multi-harmonic Web Audio.

Detailed Release Notes: Changes Since Midterm MVP

New Features Implemented:

- **Three-State Cohesive Game Loop:** Start Menu circuit selector, in-race Pause Modal (Esc) with controls guide, and End-State Checkered Flag podium summary with official lap comparison tables.
- **Manual WASD Driving & Reverse:** Full user driving controls with progressive threshold braking transitioning smoothly into reverse (-18 m/s). Front steering wheels and cockpit steering wheel turn with natural yaw inertia.
- **3D Steering Wheel Digital Dashboard:** Working physical display mounted directly on the wheel showing live gears (1-8), engine RPM shift LEDs, digital speedometer, and live lap delta timer.
- **Procedural Track Length Scalability (1,000m - 10,000m):** Live slider and numeric inputs allow users to resize circuits. Includes official dimensions for Monza, Silverstone, Spa, Monaco, and Suzuka.
- **Procedural Web Audio Engine:** Real-time additive synthesizers generating authentic F1 engine acoustics modulated by throttle and gear ratios without pre-recorded audio files.

Bug Fixes & Refinements:

- **Reverse-to-Forward Math/NaN Crash Fix:** Fixed an issue where transitioning from reverse to forward evaluated fractional powers of negative floats in Math.pow(normRatio, 1.15). Clamped acceleration ratios to non-negative domains, eliminating NaN physics crashes.
- **Steering Key Inversion Resolved:** Corrected steering input logic so 'A' turns left and 'D' turns right, matching chassis yaw and front wheel group visual orientations.
- **Dynamic Asset Placement & Clearance:** updateEnvironmentForTrack() dynamically repositions mountain rings, outer pine forests, and 16 infield groves, guaranteeing at least 22m clearance from track asphalt. Grandstands are guaranteed >=6.0m buffer outside curbs.
- **Z-Fighting & Screen Flickering Elimination:** Added logarithmicDepthBuffer with fallback, near-plane adjustment (0.6m), negative polygon offsets on road/curbs, and removed a rogue 11ms interval timer to ensure display V-Sync locked at 90 FPS.

Known Issues & Active Limitations:

- **Soft Boundary Restitution:** Off-track collisions currently apply elastic soft pushback rather than rigid impulse damage (scheduled for Week 11).
- **Planar Tire Friction on Extreme Banking:** Dynamic tire model assumes horizontal road projections, causing minor slip underestimation on vertical banking angles exceeding 15 degrees.